



HERA

HYBRID ELECTRIC REGIONAL AIRCRAFT

Group Design Project
Flight Mechanics, Performance and Control

Student: Norberto Goizueta
CID: 00824286

Project Coordinator: Dr. Errikos Levis
Academic Supervisor: Dr. Andrew Wynn
Department: Department of Aeronautics
Module: AE3-306 Group Design Project
Academic year: 2015/2016

Personal Tutor: Prof. Emile Greenhalgh
Date: 13 June 2016

Department of Aeronautics
South Kensington Campus
Imperial College London
London SW7 2AZ
U.K.



Abstract

A detailed overview of the power requirements and initial wing sizing of a hybrid-electric regional aircraft concept (HERA) has been performed in this technical report, focusing on the impact mission requirements have on the initial design.

The conceptual design phase is continued with a detailed analysis of the static stability requirements, control power and performance, as stated by the applicable civil regulations in FAR §25.

The current design stage is the end of a first iteration of the conceptual and preliminary design phase. Performance results are as of now not compliant with mission requirements but reasons for this have been identified and actions to mitigate their effect are planned for the next design iteration.

Keywords: Hybrid, Electric, Regional, Initial Sizing, Constraint Diagram, Stability, Static Stability, Control, Performance, Flight Envelope.

Foreword

This technical report is part of a compendium of data and other reports by fellow HERA members as stated in the References section. It is suggested that these reports are available to the reader as there is heavy cross-referencing between them and substantial parameters and information has derived from other HERA Divisions.

Finally, as with any aviation related document, acronyms are heavily employed. These acronyms are included in the Nomenclature and are consistent with those typically used by aviation bodies.

Acknowledgements

The author would like to acknowledge the invaluable help and advise provided by the Project Coordinator, Dr. Errikos Levis; without his advise, teachings and expertise (and endless consultation hours), this design would not have reached its current stage and would have fallen with major shortcomings. Thank you.

The HERA team need also be acknowledged for this great design over the past weeks and for always providing top-notch findings and data. This team greatest success has definitely been the group spirit throughout.

Finally, the author would like to thank the academic supervisors who have been following the progress of the project, always there to help and provide useful engineering and technical advice.

Contents

List of Figures	iii
List of Tables	iii
HERA Team Members	iv
Nomenclature	v
1 Introduction	1
2 Initial Sizing	1
2.1 Constraint Diagram Analysis	1
3 Weight and Balance	3
4 Longitudinal Controllability and Trim	4
5 Longitudinal Static Stability	5
5.1 Wing and Landing Gear Placement	7
5.2 Power Effects	8
5.3 Strut effect on Longitudinal Static Stability	8
5.4 Longitudinal Static Stability Analysis Conclusions	9
6 Lateral-Directional Static Stability	9
7 Lateral-Directional Control	10
8 Performance	11
8.1 Excess Power Plots	11
8.2 Balanced Field Length	12
8.3 Performance Conclusions	12
References	13
A Phases of Flight Evaluated in the Constraint Diagram	14
B Weight and Balance Detailed Systems Weights	15
C Preliminary Tailplane Design	16

List of Figures

2.1	Constraint Diagram for the Strut-Braced Wing Design. The takeoff condition is very demanding although it represents a minimal time fraction of the total flight. Sizing fuel engines to the remaining phases of flight will make them smaller and more efficient.	2
2.2	Variation in the ideal design point with Cruise Mach number.	3
2.3	Variation in the ideal design point with Stall Speed.	3
4.1	Free body diagram of forces and moments acting on the aircraft in stability axes. Power effects are not included in the power off static margin calculations.	5
4.2	Trim Envelope for clean aircraft at cruise at different power settings. Note the decrease in static stability in the power on case. Power has a destabilising effect.	5
4.3	Trim Envelope for the aircraft in the terminal environment at different power settings. Note the decrease in static stability in the power on case. Power has a destabilising effect.	6
5.1	Static Margin and Induced Drag at trim variation with tailplane aspect ratio and wing location.	8
8.1	Excess Power Contours for Level and Turning Flight with the flight envelope delimited by stall speed and $P_s = 0 \text{ ms}^{-1}$ line.	11
8.2	Balanced Field Length Performance.	12
C.1	Preliminary Tailplane Sizing Method.	16

List of Tables

2.1	Initial Sizing Conclusions.	3
3.1	Weight and Balance: Results and Evolution along design stage. All distances taken from the aircraft's nose.	4
4.1	Elevator Trim in different flight conditions.	6
4.2	Elevator Dimensions and Properties.	6
5.1	Aerodynamic properties of the wing and fuselage [1, 2].	7
5.2	Evolution of wing position from a stability perspective.	8
5.3	Propeller Effects on Longitudinal Stability and Trim.	9
5.4	Horizontal and Vertical Stabiliser Parameters. Final Dimensions after Stability Analysis.	9
5.5	Longitudinal Static Stability Analysis Concluding Results.	9
6.1	Lateral - Directional Stability Contributions and Results. Values for commercial aircraft have been added for comparison.	10
7.1	Absolute control surface deflection required for FAR §25.147 lateral-directional control in asymmetric thrust conditions.	11
7.2	Lateral-Directional Control Surface Sizing and Deflection.	11
8.1	Balanced Field Length Performance Results.	12
8.2	Takeoff V speeds.	12
A.1	Flight Conditions Evaluated.	14
B.1	Class II Detailed Weight and Balance break down. X reference from the nose, Z reference from 1m below fuselage.	15

HERA Team Members

A.G.	Andrew Olson Gallardo	Project Co-ordination and CAD
H.Y.	Han Yan	Project Co-ordination and CAD
A.P.	Alan Perfect	Aerodynamics
H.K.	Hong Kim	Aerodynamics
W.T.	William Taylor	Aerodynamics
H.M.	Hasan Mansoor	Flight Mechanics, Performance and Control
J.S.	Jamell Ivan Samuels	Flight Mechanics, Performance and Control
N.G.	Norberto Goizueta	Flight Mechanics, Performance and Control
F.P.	Faris Pleho	Flight Simulation
N.T.	Nigvi Tam	Flight Simulation
J.M.	James McCullagh	Structural Design
R.N.	Ryan Ng	Structural Design
S.R.	Sebastian Reymert	Structural Design
A.W.	Ashleigh Waldron	Systems and Propulsion
F.M.	Fraz Malik	Systems and Propulsion
J.K.	Julia Kasiborska	Systems and Propulsion
K.T.	Kah Ming Tan	Systems and Propulsion
S.V.	Sander James Visser	Systems and Propulsion

Nomenclature

Following is a nomenclature of scientific symbols and units used hereinafter in this document. Where a symbol is repeated, the context of the equation will suggest as to what it refers to.

General Symbols

C_{D0}	Zero-lift Drag Coefficient	— — —
C_L	Lift Coefficient	— — —
C_l	2D Sectional Lift Coefficient	— — —
C_M	Pitching Moment Coefficient	— — —
D	Drag	N
L	Lift	N
L	Rolling Moment	N m
M	Mach Number	— — —
M	Pitching Moment	N m
N	Yawing Moment	N m
P	Power	W
S	Area	m ²
V	Velocity	ms ⁻¹
W	Weight	N
X	Absolute distance along longitudinal axis measured from the nose	m
$\bar{X}$	Distance along longitudinal axis as fraction of MAC measured from the nose	— — —
b	Span (wing tip to wing tip)	m
c	Chord	m
$\bar{c}$	Mean Aerodynamic Chord	m
e	Oswald Efficiency Factor	— — —
h	Altitude	m
n	Load Factor	— — —
q	Dynamic Pressure	Pa
t	Time	s
x	Coordinate along longitudinal axis. Origin at nose.	m
y	Coordinate along lateral axis. Origin fuselage centreline	m
z	Coordinate along vertical axis. Origin at fuselage centreline	m

Greek Symbols

Γ	Wing Dihedral Angle (Positive Up)	deg
Λ	Wing Sweep Angle (Backward sweep positive)	deg
α	Angle of Attack	rad
α	Power Lapse Ratio with Altitude	— — —
β	Sideslip Angle	rad
β	Segment Weight to Takeoff Weight Ratio	— — —
γ	Flight Path Angle or Pitch	rad
δ	Deflection angle	rad
ε	Induced Downwash Angle	rad
η	Efficiency	— — —
ρ	Air Density	kg m ⁻³
$\mathcal{R}$	Aspect Ratio	— — —

Subscripts

CG	Centre of Gravity
T	Thrust
TO	Takeoff conditions
ac	Aerodynamic centre
acf	Refers to the whole aircraft
asym	Asymmetric conditions
c4	Quarter-chord position
e	Elevator
f	Refers to the fuselage
h	Refers to the horizontal stabiliser
md	Minimum Drag
max	Maximum
np	Neutral Point
prop	Propeller or propulsive
ref	Landing Reference Speed or Reference Location
v	Refers to the vertical stabiliser
w	Refers to the main wing
wf	Refers to the main wing and fuselage contribution

V speeds

V_{ef}	Critical Engine Failure Speed	kts or m/s
V_r	Rotation Speed	kts or m/s
V_{ref}	Landing Reference Speed	kts or m/s
V_s	Clean Stall Speed	kts or m/s
V_{s0}	Stall Speed in Landing Configuration	kts or m/s
V_{s1}	Stall Speed in Takeoff Configuration. Power ON.	kts or m/s
V_1	Decision Speed	kts or m/s
V_2	Climb to 35ft Speed	kts or m/s

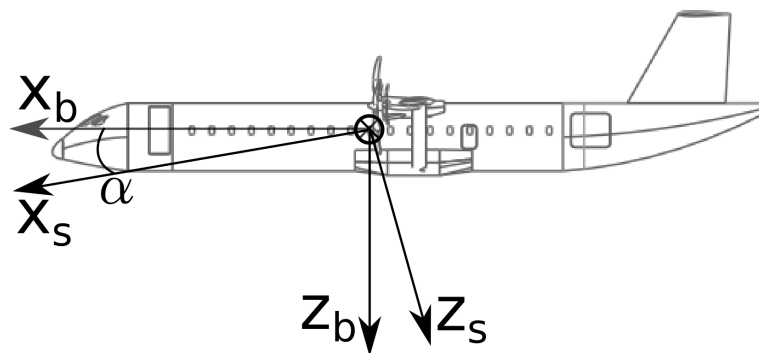
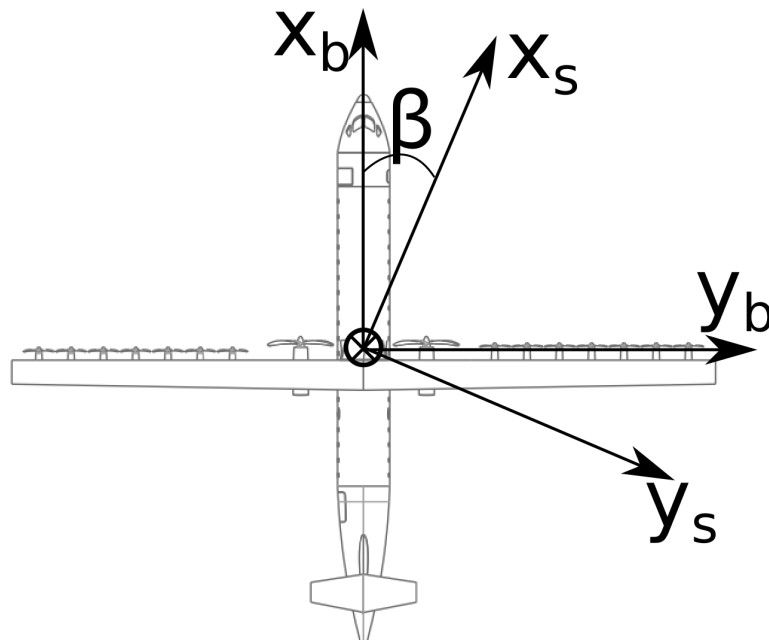
Acronyms

AEO	All Engines Operative
AFT	Aftwards
APP	Approach
ASDA	Accelerate-Stop Distance Available
ASDR	Accelerate-Stop Distance Required
BAG	Baggage
BOC	Base of Climb
CG	Centre of Gravity
CRS	Cruise
FL	Flight level (Altitude in hundreds of feet)
FAA	Federal Aviation Administration
FAR	Federal Aviation Regulations
FWD	Forward
HERA	Hybrid-Electric Regional Aircraft
HLD	High Lift Device
LE	Leading Edge
LG	Landing Gear
LDA	Landing Distance Available

LDG	Landing
MAC	Mean Aerodynamic Chord
NA	Not Applicable
PAX	Passengers
OEO	One Engine Operative
TO	Takeoff
TOC	Top of Climb
TOP	Takeoff Parameter
TODA	Takeoff Distance Available
TODR	Takeoff Distance Required
TORA	Takeoff Runway Available
VV	Vertical Velocity

Body and Stability Axes

The following Figures define body and stability axes placed on the aircraft coordinate system. These will be referenced in the stability and control section. All aerodynamic forces and moments are positive following the coordinate system convention of these axes.



1 Introduction

The aim of this technical report is to present, in a comprehensive manner, the tasks carried out by the Flight Mechanics Division of HERA in ensuring that the aircraft complied with the stability, control and performance regulations stated in FAR §25. Performance in particular was analysed to determine how to vary the design in future iterations to meet desired requirements such as takeoff distance and cruise speed.

Special emphasis has been put on control surface design, working closely with the Flight Testing Division of HERA in assuring that not only regulations are met but also that the aircraft feels comfortable and sufficiently manoeuvrable to the pilot.

Furthermore, a detailed sensitivity analysis was performed on the initial sizing process regarding the impact of design parameters such as cruise speed on the final power and wing loading of the aircraft.

For preliminary mission design, mission statement and initial baseline planning, refer to [3].

2 Initial Sizing

The initial sizing of HERA commenced after having decided the baseline configuration of the aircraft. The constraint diagram analysis was crucial in determining the advantage of the strut-braced wing aircraft over other considered configurations.

The initial sizing consists of two processes: weight and fuel estimations and power and wing loading requirements for different flight conditions. The latter will be plotted on a constraint diagram and analysed in detail. For a detailed overview of the weight estimation analysis refer to [4].

2.1 Constraint Diagram Analysis

The constraint diagram is an extremely useful tool as it shows the designer the most restrictive and demanding phases of flight in terms of wing loading and power to weight requirements. The most restrictive constraint along with the weight estimation will be the setting stone for the power plant and wing designs.

The constraint line equations are derived from the excess thrust equation shown in Equation 2.1, which can be converted to power and expanded into Equation 2.2, which can then adapted to the specific conditions of each phase. As shown, zero-lift drag and aspect ratio play a crucial role in determining the needed power for flight. Based on geometry data from the baseline configuration, preliminary aerodynamic coefficient estimations can be made, such as the Oswald efficiency factor, estimated as $e \approx 0.8$ [1, 5]. Legal climbing requirements during single-engine operations as stated by FAR §25.111 are also included and are stated in Table A.1. Aircraft geometrical properties can be referenced from the HERA Specification Sheet.

$$\frac{dh}{dx} = \frac{T - D}{W} \quad (2.1)$$

$$\left. \frac{P}{W} \right|_{TO} = \frac{V}{\eta_{prop}\alpha} \left(\frac{qC_{D0}}{W_{TO}/S} + \frac{1}{\pi Re} \frac{n^2 \beta^2 W_{TO}}{q S} \right) + \frac{\beta}{\alpha} \left(\frac{1}{V} \frac{dh}{dt} + \frac{1}{g} \frac{dV}{dt} \right) \quad (2.2)$$

The power to weight and wing loading lines for different flight phases are shown in Figure 2.1, the constraint diagram. The selected design point will satisfy all constraints and lie to the left of the stall and landing lines. The ideal design is that which has the lowest power to weight requirements and largest wing loading. For an aircraft with a single energy source, there exists only one design point that determines the powerplant sizing. However, having HERA two sources of energy (fuel and electric), the design is far more flexible. As seen on the diagram, takeoff imposes a very stringent constraint on the power requirement compared to the other phases of flight at the same wing loading. This reason enables the hybrid design to be fully exploited. The total power produced by both types of engines can be sized according to takeoff requirements [4400N/m², 26W/N] yet the beauty lies in that the fuel powered engines can then be sized for the other less stringent flight conditions [4400N/m², 20W/N].

This means that the electric engines will only have to provide a supplemental power boost for takeoff and that sufficient power will be provided solely by the fuel engines for the remainder of the flight. The advantage of the hybrid design is therefore twofold: the fuel engines are downsized compared to a conventional design for the same mission profile and the fuel engines will operate at nearly 100% power during cruise, which is the most efficient operating region for a turboprop engine [6]. A sensitivity analysis on the parameters affecting the

constraint diagram helped determine the most optimal operating conditions and design aims like optimal cruise speed, stall speed.

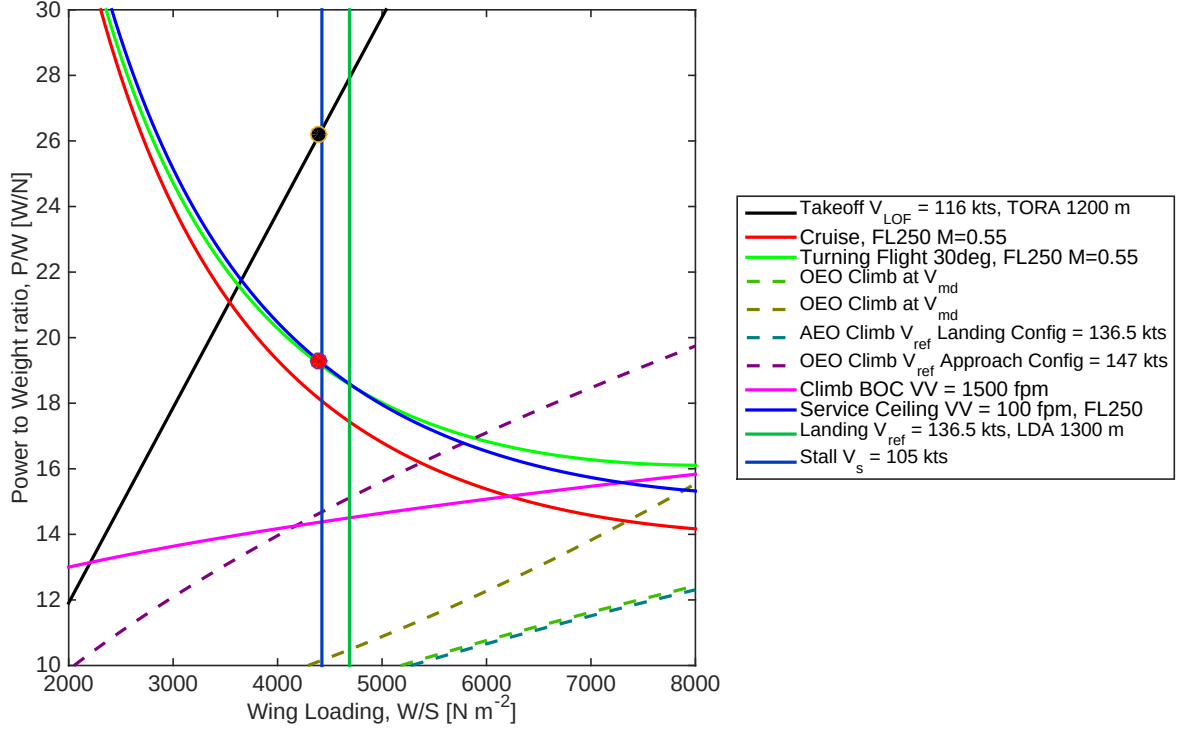
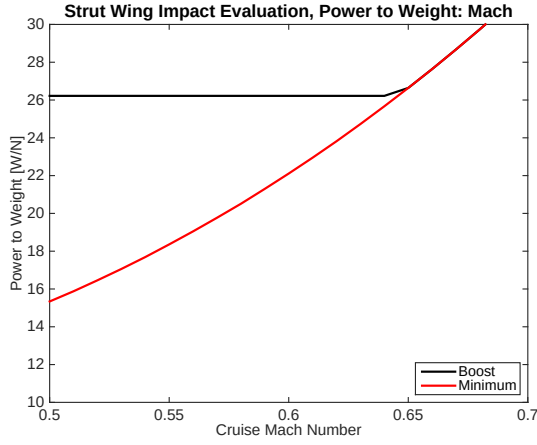


Figure 2.1: Constraint Diagram for the Strut-Braced Wing Design. The takeoff condition is very demanding although it represents a minimal time fraction of the total flight. Sizing fuel engines to the remaining phases of flight will make them smaller and more efficient.

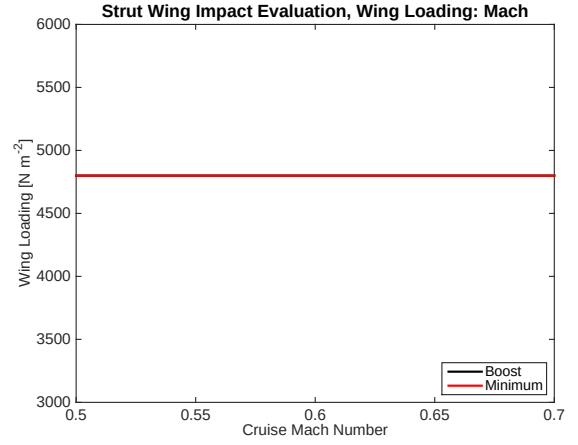
The variation of the design point with cruise Mach number is shown in Figure 2.2. Wing loading is not affected as the limiting constraint is the stall line, as shown in Figure 2.2b. It can be seen that at higher Mach numbers the power requirement during cruise increases significantly to the point that cruise becomes the limiting constraint above $M = 0.65$, eliminating the need for an electric boost during takeoff. Although it seems beneficial to fly slower from a design standpoint, it is in the best interest to cruise as fast as possible. Given that the current cruise speeds of conventional regional turboprops lie around $M \approx 0.45$, HERA would be at a competitive advantage if a cruise speed around $M = 0.5$ were achieved. Following this cruise condition, the electric boost required for takeoff ($\approx 2.1\text{MW}$) was, however, deemed unacceptably high for the electric motors [7]. Therefore, the fuel engines were sized up to provide enough power for cruise at $M = 0.55$ and in that way reduce the electric boost requirement to approximately 1.6MW .

Figure 2.3 shows the variation in the design point with stall speed. Regarding power to weight, Figure 2.3a shows the divergence between the fuel engine sizing and the required electric boost as stall speed increased. Similarly, wing loading also increases with stall speed (Figure 2.3b). This suggests that it is beneficial to increase stall speed in terms of sizing. However, manoeuvrability and approach speeds must be taken into account. A too high stall speed increases the approach category of the aircraft, limiting the number and type of instrument approaches that can be performed in poor meteorological conditions. Therefore, it was chosen to cap stall speed at 105 kts, upper limit of the Approach Category C (FAR §97.3), the most common for turboprop regional aircraft.

Therefore, from the mission design parameters' impact on the constraint diagram along side with the weight estimations from [4], the initial sizing process can be concluded. Results are shown in Table 2.1 for takeoff (fuel and electric power) and cruise (fuel power only). Data for a conventional turboprop is provided for reference.

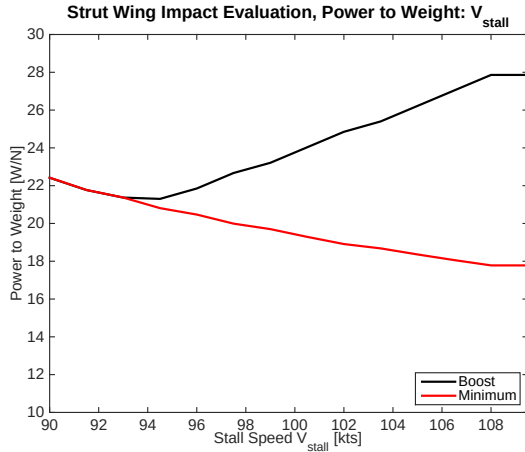


(a) Variation of the design point (Power to Weight) with increasing Mach number. The electric boost advantage for takeoff can be observed to diminish with increasing Mach.

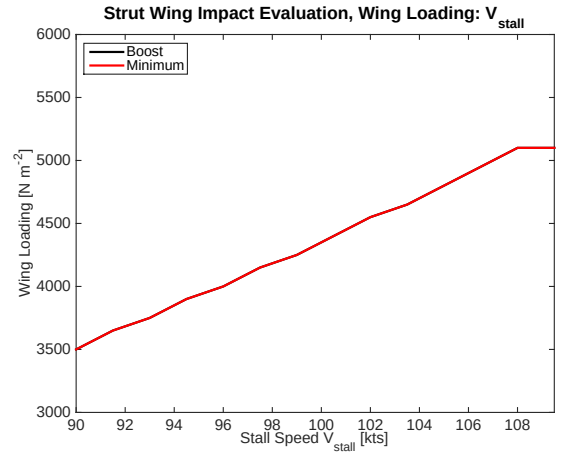


(b) Variation of the design point (Wing Loading) with Mach number. Given the stall speed constraint, there is no variation in wing loading with Mach number.

Figure 2.2: Variation in the ideal design point with Cruise Mach number.



(a) Variation of the design point (Power to Weight) with increasing Stall speed. The electric boost range is increased as the stall speed is increased.



(b) Variation of the design point (Wing Loading) with stall speed. A beneficial increase in W/S with stall speed is observed.

Figure 2.3: Variation in the ideal design point with Stall Speed.

Table 2.1: Initial Sizing Conclusions.

	Power To Weight [W/N]	Wing Loading [N/m ²]	Weight [kg]	Power [MW]	Wing Area [m ²]	Cruise Mach
HERA (TO)	26.2	4400	21 500	5.6	48	0.55
HERA (CRS)	19.3	4400	21 500	4.0	48	0.55
Conventional Turboprop [8]	16.5	3700	23 000	3.7	61	0.4

3 Weight and Balance

The Weight and Balance estimation method as in [9] was based on two types of data. Reasonable estimates of structural and system weights made by the different HERA divisions and empirical methods based on historical data developed by [10] for those aircraft systems with unknown weights at this stage.

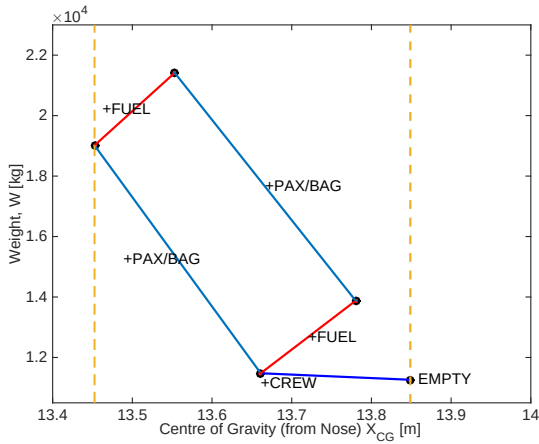
Passenger and Baggage weight estimates have been made using survey data provided by airlines for the regional category aircraft and includes statistical data on the number of checked baggage as well as weight per

passenger to include carry on items [11].

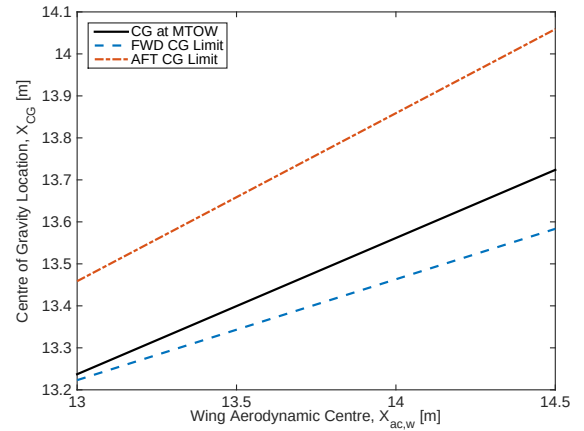
Different loading conditions have been considered as shown in the CG envelope in Figure 3.1a, which outlines HERA's final weight distribution. These include empty weight, operating empty weight (crew and trapped fuel and oil) and the different possible payload and fuel arrangements. Final results are tabulated in Table 3.1, which shows the evolution of the weight estimates as the design progressed and, as it can be seen, the results are consistent and convergent. A more detailed breakdown of the weight and balance can be found in Appendix B.

Table 3.1: Weight and Balance: Results and Evolution along design stage. All distances taken from the aircraft's nose.

Design Stage	MTOW [kg]	Empty Weight [kg]	FWD CG Limit [m]	AFT CG Limit [m]
Initial Sizing	21 500	NA	NA	NA
Class I	21 000	13 400	13.3	13.6
Class II	21 500	11 300	13.5	13.9



(a) Class II Weight and Balance Envelope for the different loading conditions of the aircraft.



(b) Variation of the Centre of Gravity Position with varying wing position.

4 Longitudinal Controllability and Trim

The longitudinal trim analysis and design of control surfaces is usually done simultaneously with the longitudinal static stability analysis performed in §5 after the preliminary tailplane sizing shown in Appendix C.

Trim condition is referred as the condition in which the aircraft is in moment equilibrium: $C_{M,ref} = 0$. Moments are taken about the reference location, in this case taken to be at the design MTOW CG $X_{CG,MTOW} = 13.6$ m. However, since the CG varies depending on fuel and payload loadings, the most aft and fore CG limits from Table 3.1 must also be analysed.

Figure 4.1 shows the aerodynamic forces and moments acting on the aircraft. Taking moments about the reference point $X_{ref} = X_{CG,MTOW}$, the trim equation in non-dimensional form is obtained as shown in Equation 4.1. The tailplane plays an important role given the large moment arm.

$$C_{M,ref} = C_{L,w}(\overline{X_{CG}} - \overline{X_{ac,w}}) + C_{M,w} + C_{M,f} + \eta_h \frac{S_h}{S_w} C_{L,h}(\overline{X_{CG}} - \overline{X_{ac,w}}) = 0 \quad (4.1)$$

For the aircraft to be able to be in trim at any flight condition, an elevator is located on the horizontal stabiliser to provide the pilot with a means of controlling pitch and trim. It is important that the tailplane and elevator are sized properly to provide controllability in all flight conditions, as it may be the case, for instance, that the maximum elevator deflection is not sufficient to trim the aircraft at low speeds. The influence of the elevator tab in the lift produced by the horizontal stabiliser is derived empirically by idealising it as a plain flap and accounting for losses in the case the hinge gap is not sealed and for separation at high deflection angles [12, 13].

To find the trim condition, a graphical method is used. Constant elevator deflection lines are plotted on a

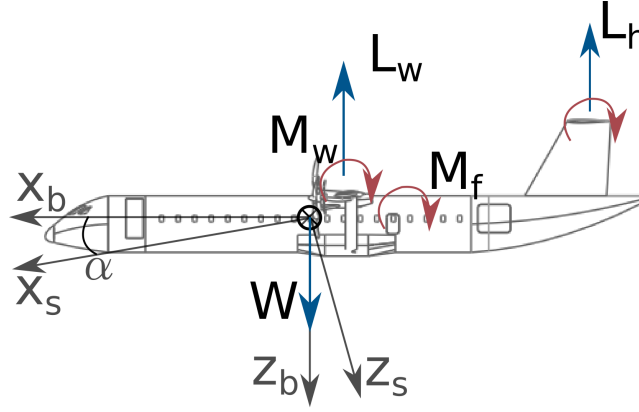
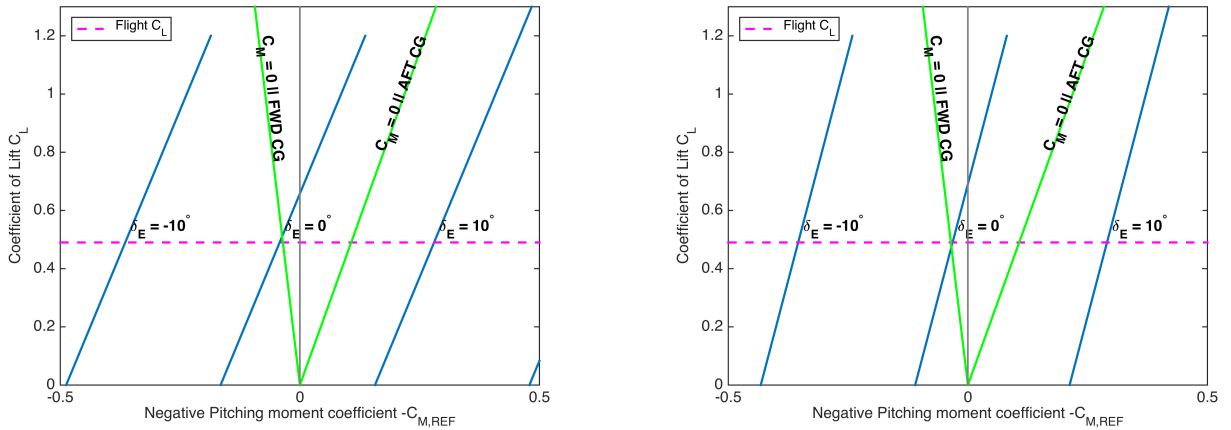


Figure 4.1: Free body diagram of forces and moments acting on the aircraft in stability axes. Power effects are not included in the power off static margin calculations.

pitching moment - lift coefficient graph. The $C_M = 0$ or trim condition can then be easily identified by the intersection of both lines, or in its defect it can be interpolated to find the required elevator deflection for trim at a certain condition. This can be seen on Figure 4.2. Note that the strict $C_M = 0$ has been taken with the CG at $X_{ref} = X_{CG,MTOW}$, but the most fore and aft limits are also shown, effectively delimiting the trim envelope of the aircraft.



(a) Power OFF Clean Configuration Trim Envelope with cruise C_L and CG limits shown.

(b) Power ON Clean Configuration Trim Envelope with cruise C_L and CG limits shown.

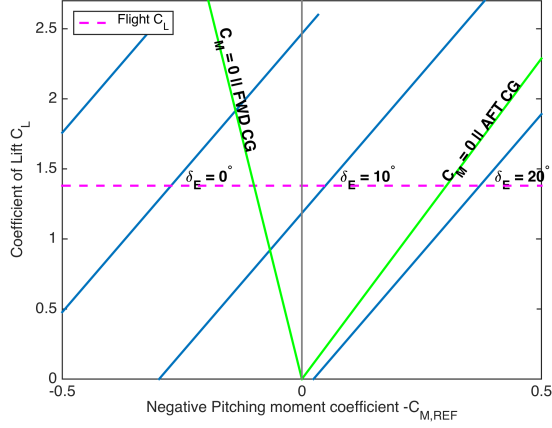
Figure 4.2: Trim Envelope for clean aircraft at cruise at different power settings. Note the decrease in static stability in the power on case. Power has a destabilising effect.

The elevator sizing process is iterative. The trim envelopes shown in Figures 4.2 and 4.3 must be referenced after any sizing changes to ensure that proper control is attainable in all phases of flight. A change in horizontal stabiliser incidence can also be used to this extent. In this case, the aim was to achieve a near zero elevator tab deflection at cruise, which required a negative tail incidence of -3° . Table 4.1 shows the trim analysis results, which determine the required deflection of the elevator tab. This trim analysis was considered acceptable and the elevator sizing obtained from it was fed to the Flight Testing division of HERA. Pilot feedback was satisfactory. The consequent elevator dimensions are shown in Table 4.2. The destabilising power effects observed in Figures 4.2 and 4.3 shown by the lines becoming steeper will be discussed in §5.

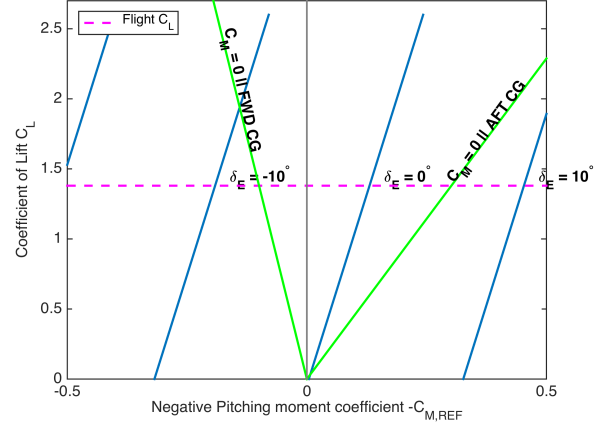
5 Longitudinal Static Stability

Given the symmetry of the aircraft about the XZ plane, the motion can be decoupled into longitudinal and lateral components. Longitudinal motion will be mostly affected by weight and the centre of gravity location, lift from lifting surfaces and pitching moments. These are shown in Figure 4.1.

An aircraft must be longitudinally statically stable per FAR §25.173, meaning that upon a disturbance that



(a) Power OFF Terminal Environment Trim Envelope with approach C_L and CG limits shown.



(b) Power ON Terminal Environment Trim Envelope with approach C_L and CG limits shown.

Figure 4.3: Trim Envelope for the aircraft in the terminal environment at different power settings. Note the decrease in static stability in the power on case. Power has a destabilising effect.

Table 4.1: Elevator Trim in different flight conditions.

	Stabiliser Incidence [deg]	Elevator at trim FWD CG [deg]	Elevator at trim MTOW CG [deg]	Elevator at trim AFT CG [deg]
Power OFF, Clean	-3	0	1	5
Power ON, Clean	-3	0	1	4
Power OFF, APP	-3	+5	+8	+18
Power ON, APP	-3	-7	-4	+5

Table 4.2: Elevator Dimensions and Properties.

Type	Chord Fraction c_e/c_h	Span Fraction b_e/b_h	Deflection Range [deg]
Plain Flap	0.25	0.9	-20 - +20

results in a change in angle of attack, a restoring moment is produced and the aircraft shows the tendency to return to the equilibrium point. FAR §25.173 states that longitudinal static stability must be achieved in all flight conditions and at all flight speeds. This criterion is mathematically expressed as:

$$\frac{\partial C_M}{\partial C_L} < 0 \quad (5.1)$$

The static margin is defined as the distance between the neutral point and the centre of gravity as shown in Equation 5.2. If the centre of gravity is coincident with the neutral point, the aircraft is neutrally stable. For stability, the static margin must be positive and is typically $SM \approx 15\%-20\%$ for commercial transport aircraft [14]. This condition forces the neutral point to be aft of the allowable CG range (For CG ranges, see §3).

$$SM = -\frac{\partial C_M}{\partial C_L} = \frac{x_{np} - x_{CG}}{\bar{c}} \quad (5.2)$$

The neutral point or aerodynamic centre of the aircraft can be calculated by summing the static moment contributions of the lifting surfaces as shown in Equation 5.3 [12]. This corresponds to the clean configuration power off neutral point; the use of HLDs and power effects will have an effect on stability that will be discussed in this section.

$$\bar{X}_{np} = \frac{\bar{X}_{ac,wf} C_{L_{\alpha,wf}} - C_{M_{\alpha,f}} + \eta_h \left(1 - \frac{\partial \epsilon}{\partial \alpha}\right) \frac{S_h}{S_w} C_{L_{\alpha,h}} \bar{X}_{ac,h}}{C_{L_{\alpha,acf}}} \quad (5.3)$$

Methods for calculating lift-curve slopes from Equation 5.3 for the main wing and entire aircraft can be found in [1] and the results are shown in Table 5.1. Fuselage effects, namely the shifting of the aerodynamic centre of the wing-fuselage combination forward ($\Delta X_{ac,wf}$) and the pitching moment effects can be found in [2].

Tailplane coefficients present in Equation 5.3 is tabulated in Table 5.4 and a more detailed overview of its design can be found in Appendix C. Note that η_h refers to the ratio of dynamic pressure of the horizontal stabiliser to that of the wing. Due to the T-tail design it has been taken as $\eta_h = 1$.

Table 5.1: Aerodynamic properties of the wing and fuselage [1, 2].

Configuration	$C_{L_{\alpha, wf}}$ [/rad]	$C_{L_{\alpha, acf}}$ [/rad]	$\Delta X_{ac, wf}$
Clean	5.52	6.03	-0.77
TO/LDG	5.98	6.22	-0.77

5.1 Wing and Landing Gear Placement

One of the Flight Mechanics Division's main tasks was to decide on wing location and tailplane sizing. As with any cross-functional design, compromises must be made so that the adopted solution is optimum for the aircraft as a whole and not best for an individual subsystem. In this case, coordination between Structures, Systems and Flight Mechanics divisions was essential.

The wing's wing box is attached to the fuselage section at the two wing spar locations with heavy frames that reinforce this high-load region. Therefore, it is intuitive to use this already reinforced fuselage section to attach the main landing gear, instead of using a separate heavy fuselage frame that may incur structural weight penalties of up to 20% of the fuselage's weight [4]. The optimal structural position will be to align the landing gear with the heavy frame that connects with the rear wing spar, located at 60% of the wing root chord [15].

Furthermore, the landing gear placement depends heavily on the centre of gravity position as the aircraft must not tip over whilst on the ground irrespective of the cargo loading and the load on the nose gear must be between 6% and 20% [16]. As expected, the final position will be obtained after several iterations and evaluations of the required compromises between divisions.

From a Stability perspective, the most favourable position is when the tailplane size is minimum for sufficient stability and control (to reduce skin friction drag) and the lift produced by the tailplane in trim at cruise is close to zero (to minimise induced drag). Figure 5.1 shows the the impact wing position and tailplane aspect ratio have on induced drag at trim and static margin, the two outputs that need careful monitoring. From Figure 5.1a it can be seen that the optimal design point to minimise induced drag at trim is with $X_{ac, w} \approx 13$ m and a high aspect ratio tailplane. Selecting a high tailplane aspect ratio would offer the flexibility of being able to move the wing without a significant drag penalty.

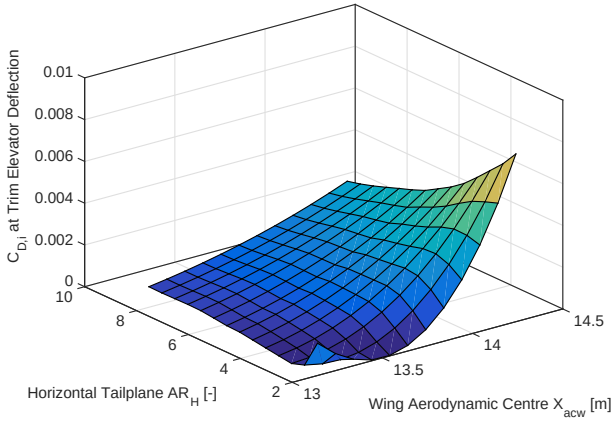
To achieve a desirable static margin of 15%-20%, Figure 5.1b shows that with a high tailplane aspect ratio, a wing location near $X_{ac, w} = 13$ m will be required. This suggests an optimum location in terms of stability: $X_{ac, w} = 13$ m. The desired SM is achieved with a minimal tailplane induced drag impact.

If this optimum point for stability is chosen, the resultant centre of gravity range lies between 13.2 m and 13.5 m from the nose, as shown in Figure 3.1b. The rear spar and hence the optimal landing gear attachment point would then lie at 13.5m which is coincident with the aft CG limit. This position is unacceptable given that the nose wheel would carry virtually no load and that any improper loading could result in the aircraft tipping back on the ground, with the consequent structural damage that would cause.

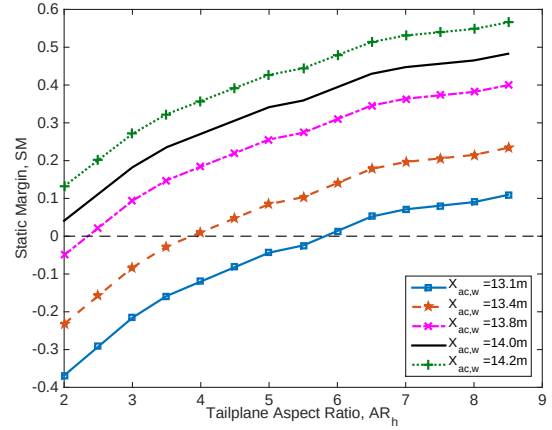
From a landing gear design point of view, it is desirable to shift the wing aft such that the aftmost CG limit lies ahead of the rear spar at a position that places an 8% weight load on the nose gear. This position translates into a wing aerodynamic centre location at 14.5m from the nose. If the aspect ratio of the tailplane is kept high (>7), the drag penalty is not substantial. However, the static margin would be unacceptably high ($>50\%$). Therefore, a compromise between functional teams must be reached.

Landing Gear relaxed their nose loading requirement from 8% to 6%, which allowed having a shorter distance between the CG and main gear. As a consequence the wing could be shifted forward [7] from their ideal point. On the other hand, Stability opted for a lower aspect ratio horizontal tailplane with the cost of an induced drag penalty. This mitigated the effect of moving the wing aft that increased the SM beyond acceptable upper limits.

After this analysis, it was decided to place the wing at $X_{ac, w} = 14$ m and with a tailplane aspect ratio $\mathcal{R}_h = 4$



(a) Variation of the induced drag in trimmed cruise with wing position and horizontal stabiliser aspect ratio.



(b) Variation of Static Margin with horizontal tailplane aspect ratio for different wing locations. $X_{ac,w} = 14\text{m}$ became the design position.

Figure 5.1: Static Margin and Induced Drag at trim variation with tailplane aspect ratio and wing location.

as it fulfilled all requirements. Results are tabulated in Table 5.2 showing the evolution from a Stability perspective; for LG perspective see [7]. For stability, the design point represents a $SM = 25\%$, slightly above optimum yet feasible and an increase in induced drag coefficient of $\Delta C_{D,i} = 0.004$, representing an increase of about 2000N from the optimal stability design point location. However, this compromise has been required to be able to place the landing gear at the most efficient structural point.

Table 5.2: Evolution of wing position from a stability perspective.

	Wing $X_{ac,w}$ [m]	Neutral Point X_{np} [m]	Tailplane AR	Static Margin SM	Induced Drag Coefficient $C_{D,i}$	Comments
Optimum Stability	13.0	13.5	9	0.2	0.001	Unacceptable LG position.
Optimum LG	14.5	14.0	2	0.2	0.01	Unacceptable Drag Increase.
Optimum LG	14.5	14.5	9	0.6	0.001	Unacceptable Static Margin.
Compromised Location	14.0	13.9	4	0.25	0.005	Acceptable LG and SM.

5.2 Power Effects

The effects of propeller power on longitudinal static stability are not negligible and should be analysed carefully to ensure the aircraft is stable at all power conditions. Propeller effects on longitudinal static stability are due to thrust-line offset and by the propeller turning normal forces. Propeller effects on longitudinal trim are due to thrust-line offset and propeller slipstream influence on the wing, as the slipstream varies the actual velocity seen by the wing and hence changes the lift distribution.

These effects are tabulated in Table 5.3. The actual derivations are not included since they are based mainly on empirical relations derived from wind tunnel tests on propellers; for reference see [12] and [13]. Regarding the effect on longitudinal static stability, power has a destabilising effect accountable for nearly 10% Static Margin, hence the power off Static Margin at 25% provides sufficient margin to maintain static stability at high power settings.

5.3 Strut effect on Longitudinal Static Stability

HERA counts with an innovative design, making use of a strut to support the wing enabling a lower thickness to chord ratio and an unprecedented ultra-high aspect ratio. This strut will have an aerodynamic effect on the

Table 5.3: Propeller Effects on Longitudinal Stability and Trim.

Effect	ΔC_M	ΔSM
Thrust-line offset	-0.06	-0.1
Propeller Slip-stream	0.005	NA
Normal Force	NA	0.01
Total Power Effects	-0.065	-0.09

aircraft's overall lift and drag distributions. However, at this stage in the design, it has been only modelled as a parasitic drag source, neglecting any lifting contributions. The effect of the drag produced by the strut on stability is negligible for two reasons: the actual drag force is small compared to other aircraft components and the vertical distance from the strut's aerodynamic centre to the centre of gravity of the aircraft is minimal, resulting in a negligible pitching moment effect due to drag.

However, qualitatively the effect of a lifting strut on stability can be analysed. The strut can be treated as a wing with an extreme dihedral and due to the small chord, its aerodynamic centre can be treated as coincident with the wing's mid chord, its attachment point. Therefore, any lift generated by the strut will have a stabilising effect, as it will push the aerodynamic centre of the aircraft further aft, similar to the tailplanes contribution in Equation 5.3. Therefore, pending a proper numerical analysis, a destabilising effect of the lifting strut on the longitudinal static margin of the aircraft is not expected, rather quite the contrary.

5.4 Longitudinal Static Stability Analysis Conclusions

After performing a detailed static stability analysis on the longitudinal motion of the aircraft, taking into account the different stabilising and destabilising effects, the results in Table 5.5 are obtained. After this stage the wing position and horizontal tailplane size and position can be considered frozen for the first iteration of the design. Furthermore, its structural integrity has been verified and deemed satisfactory [17].

Table 5.4: Horizontal and Vertical Stabiliser Parameters. Final Dimensions after Stability Analysis.

	Area [m ²]	Taper Ratio	Span [m]	Root Chord [m]	MAC [m]	Sweep LE [deg]	$C_{L_{\alpha,h}}$	$\mathcal{R}$	$\partial\epsilon/\partial\alpha$
Horizontal Stab.	7.0	0.38	5.29	1.91	1.41	5	3.9	4	0.11
Vertical Stab.	9.28	0.53	3.42	3.77	2.97	35	1.5	1.3	-

Table 5.5: Longitudinal Static Stability Analysis Concluding Results.

	$X_{ac,w}$ [m]	X_{CG} [m]	X_{np} [m]	SM
Power OFF	14.0	13.6	13.9	0.25
Power ON	14.0	13.6	13.8	0.15

6 Lateral-Directional Static Stability

FAR §25.177 states the requirements in terms of lateral and directional stability. The former refers to raising the lower wing in sideslip with the ailerons free (Equation 6.1) whilst directional stability refers to the tendency to recover from a skid with the rudder free (Equation 6.2).

$$\frac{\partial L}{\partial \beta} < 0 \quad (6.1)$$

$$\frac{\partial N}{\partial \beta} > 0 \quad (6.2)$$

Regarding lateral stability, the main contributors to the rolling moment are the wing and the wing-fuselage interference effects [18]. For wing effects, given the ultra-high wing $\mathcal{R}$ strip theory can be used: 2D effects are calculated on an infinitesimally thin strip along the wing chord and then integrated along the span for the total effect. The main effects are the dihedral and the effective sweep effects. For the latter, given the zero sweep of the wing, the effect is zero as in a sideslip condition the effect due to the apparent backsweep of one wing is cancelled by the apparent frontsweep of the other. The wing's contribution due to dihedral to the rolling

moment derivative with respect to sideslip is shown in Equation 6.3 [18]. As shown, positive dihedral has a stabilising effect.

$$\left. \frac{\partial L}{\partial \beta} \right|_w = -\frac{2\Gamma}{S_w b_w} \int_0^{b_w/2} \frac{dC_l}{d\alpha} c_y dy \quad (6.3)$$

The lateral static stability requirement is met and results are tabulated in Table 6.1. Given the ultra-high aspect ratio, these results cannot be validated with the conventional methods in [13]; although it can be ascertained that the stability derivatives are of the correct order of magnitude by comparing them to current commercial airliners [12]. Incorporating the feedback from the Flight Testing Division for the zero dihedral case, a dihedral $\Gamma = 1^\circ$ was added to provide greater lateral stability and a more comfortable ride.

Regarding directional stability, the main contributions arise from the fuselage and the vertical tailplane. Calculations of the individual contributions are found in [13] and shown in Table 6.1. Although the FARs simply require positive directional stability, the minimum vertical stabiliser that achieved it did not offer a comfortable ride when tested by the Flight Testing Division. Therefore, the area was increased based on pilot feedback to improve control and feel of the aircraft.

Table 6.1: Lateral - Directional Stability Contributions and Results. Values for commercial aircraft have been added for comparison.

	HERA		Boeing 747		King Air 200	
	C_{L_β}	C_{N_β}	C_{L_β}	C_{N_β}	C_{L_β}	C_{N_β}
Wing + Fuselage	-0.11	-0.05	-	-	-	-
Horizontal Stabiliser	-0.08	-	-	-	-	-
Vertical Stabiliser	-0.02	0.16	-	-	-	-
Combined	-0.21	-0.11	-0.281	0.184	-0.130	0.120

7 Lateral-Directional Control

Lateral-directional control is concerned with the ability to yaw into the operative engine and to roll with and against the inoperative engine such that normal manoeuvres can be executed during asymmetric thrust operations. Motion about the two axes is coupled: there is a roll due to yaw and vice versa; reasons for this coupling will be explained in this section.

The ailerons and rudder must be sized so achieve the required manoeuvrability specified in FAR §25.147. Again, this becomes an iterative process as it is not sufficient to comply with the legal minimums but also it is important to offer a proper controllability to pilots.

From the static lateral equations of motion (derived in appendix), a linear system can be approximated and solved for the specified flight conditions. The required absolute deflections of control surfaces to can be used to size the control surfaces. Equation 7.1 shows the system of static lateral-directional equations of motion [19].

$$\begin{bmatrix} C_{Y_\beta} & C_{Y_{\delta_a}} & C_{Y_{\delta_r}} \\ C_{L_\beta} & C_{L_{\delta_a}} & C_{L_{\delta_r}} \\ C_{N_\beta} & C_{N_{\delta_a}} & C_{N_{\delta_r}} \end{bmatrix} \begin{bmatrix} \beta \\ \delta_a \\ \delta_r \end{bmatrix} = \begin{bmatrix} -C_L \sin(\phi) \cos \gamma \\ -L_{T1}/q/S_w/b_w \\ -N_T/q/S_w/b_w \end{bmatrix} \quad (7.1)$$

As seen in Equation 7.1, there are coupling effects between motions. For instance, $C_{N_{\delta_a}}$ is yaw due to aileron deflection, or adverse yaw: a differential aileron deflection will cause the upper wing to produce a higher induced drag (higher lift wing) compared to the lower wing, generating a yaw in the opposite direction to the roll. Coupling effects due to ailerons have been calculated using the strip theory method mentioned in §6. These effects are expected to be more prominent than in conventional aircraft due to the large moment arm given the aircraft's span. Remaining derivatives have been calculated following [13], given that these components are much more conventional and they fit the experimental methods [13] is based on.

Results presented in Table 7.1 show the deflection ranges needed for the required lateral control in asymmetric thrust conditions. FAR §25.147 states that the aircraft must be controllable with the two critical engines inoperative. Given the unconventionality of HERA's design, there isn't a specific FAR requirement, but interpretation of the regulations must be made. Having interpreted the spirit of the regulation, the HERA Legal Division believe that HERA qualifies as a 4+ engine aircraft during takeoff but as a twin engine during climb and

cruise. Therefore, both aspects of the regulation must be complied with. For the 4+ engine aircraft case, FAR §25.147 states the case that the two critical engines fail: following the propulsive system layout diagram [20], this corresponds to one turboprop engine and the outboard-most electrical engine of that same wing. For the twin engine aircraft case during climb, it will be the failure of a single turboprop that is considered. Therefore, after the iterative process for aileron and rudder deflection was performed and following the legal requirements and incorporating pilot feedback, the final control surface sizing is presented in Table 7.2.

Table 7.1: Absolute control surface deflection required for FAR §25.147 lateral-directional control in asymmetric thrust conditions.

	Legal Category	$ \delta_a $	$ \delta_r $
Cruise Config.	Twin	11.0	0.6
Approach Config.	Twin	22.5	0.9
Takeoff Config.	4+	21.7	0.8

Table 7.2: Lateral-Directional Control Surface Sizing and Deflection.

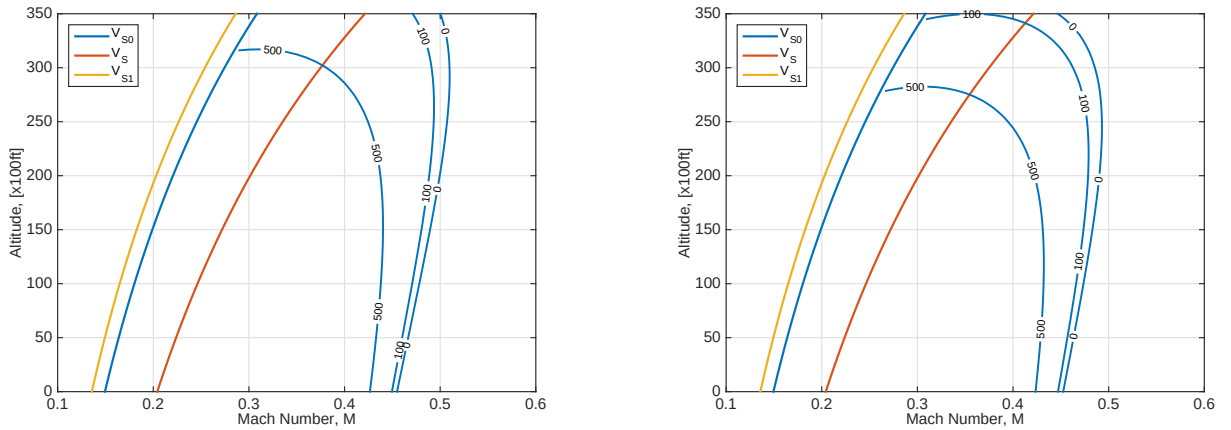
	Inboard Spanwise Location	Outboard Spanwise Location	Chord Fraction	Deflection Range [deg]
Ailerons	65%	85%	0.25	-25 – +25
Rudder	5%	95%	0.25	-20 – +25

8 Performance

Aircraft performance revealed the true characteristics of HERA. Note that although not all the mission requirements are met, these tools are useful in knowing what parameters need to be varied for the second aircraft iteration. No single aircraft has ever been built from scratch in one go.

8.1 Excess Power Plots

Although they are more commonly used for fighters, the excess power plot is an intuitive and graphical way of visualising the aircraft’s operating envelope and identifying what design aspects need modification. Focusing on



(a) Excess power contours at MTOW for steady level flight with $n=1$.

(b) Excess power contours at MTOW with a constant altitude 30degree turn $n=1.15$.

Figure 8.1: Excess Power Contours for Level and Turning Flight with the flight envelope delimited by stall speed and $P_s = 0 \text{ ms}^{-1}$ line.

the steady level flight excess power plot in Figure 8.1a, it can be seen that the desired cruise performance $M = 0.55$ at FL 230 cannot be achieved as HERA is only able to achieve $M = 0.51$ at that altitude. The reason for these discrepancies between the design statement and the performance output is a higher than expected drag contribution due to turbulent flow over the entire wing span due to propeller wash [1]. Compared with the

initial drag estimates for the initial sizing process, there has been a 24% increase in zero-lift drag, proving that underperformance should not be unexpected. Should less wingspan be in the direct propeller wash in further iterations, drag reductions of at least 10% are expected [1].

Figure 8.1b shows the same trend, indicating that at its current aerodynamic shape, the aircraft needs more power to comfortably perform a turn at cruise altitude at $M = 0.5$. Furthermore, the absolute service ceiling above FL 350 lies substantially above the design ceiling of FL 250, in part due to the high lift to drag ratio $L/D \approx 23$ at low speeds [1]. This data has been confirmed by HERA’s Flight Testing Division and an in-depth comparison of theoretical versus experimental data can be found in [21, 22].

8.2 Balanced Field Length

The balanced field length is defined when the distance required to accelerate to V_{ef} and continue takeoff to reach 35ft at the end of the clearway equals the distance required to accelerate to V_{ef} and the required distance to bring the aircraft to a halt. The graphical depiction of engine failure speed V_{ef} versus TODR or ASDR with an engine failure at that speed is shown in Figure 8.2.

The intersection of the TODR and ASDR lines represents the balanced field length and the speed at which that occurs is the critical engine speed V_{ef} . A reaction time of 2s is then added to account for the pilot acknowledging the failure and taking appropriate action per FAR §25.109. This speed becomes the decision speed V_1 .

FAR §25.113 references the aircraft’s takeoff distances and the certified distance for a particular operation becomes the greater of the balanced field length or the 115% of the normal operations takeoff distance. The normal operations takeoff distance can be found in [3] and results are tabulated in Table 8.1.

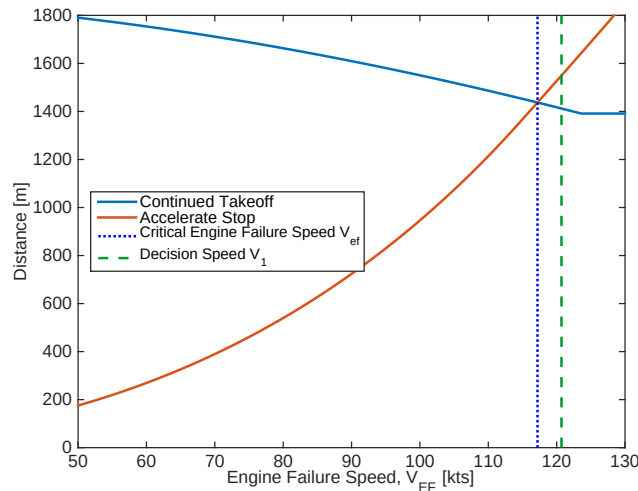


Figure 8.2: Balanced Field Length Performance.

Table 8.1: Balanced Field Length Performance Results.

Balanced Field Length	1437 m
Takeoff Distance Required OEO	1320 m
Certified TO Distance	1520 m

Table 8.2: Takeoff V speeds.

Speed		kts	ms^{-1}
Critical Engine Failure	V_{ef}	117	60
Decision	V_1	121	62
Rotation	V_r	123	63
Climb-out	V_2	128	65

8.3 Performance Conclusions

Having performed an excess power analysis and a balanced field length analysis several conclusions can be made for the next iteration in HERA’s design. It is clear from the two analyses that HERA does not comply with two of its mission requirements: cruise speed and takeoff distance. As previously mentioned, the increased drag is an important reason for it but the HERA team have already identified measures to mitigate this drag in order to comply with the mission statement in further iterations. However, as a first pass over the aircraft’s conceptual design, results are satisfactory as it will not take major design changes to adapt the current HERA design to one that fully complies with the specification.

References

- [1] Kim HY. Hybrid Electric Regional Aircraft - Aerodynamics. Department of Aeronautics, Imperial College London; 2016.
- [2] Perfect A. Hybrid Electric Regional Aircraft - Aerodynamics. Department of Aeronautics, Imperial College London; 2016.
- [3] Olson-Gallardo A. Hybrid Electric Regional Aircraft - Project Management. Department of Aeronautics, Imperial College London; 2016.
- [4] Reymert S. Hybrid Electric Regional Aircraft - Structural Design. Department of Aeronautics, Imperial College London; 2016.
- [5] Taylor W. Hybrid Electric Regional Aircraft - Aerodynamics. Department of Aeronautics, Imperial College London; 2016.
- [6] Tan KM. Hybrid Electric Regional Aircraft - Propulsion and Systems. Department of Aeronautics, Imperial College London; 2016.
- [7] Visser SJ. Hybrid Electric Regional Aircraft - Propulsion and Systems. Department of Aeronautics, Imperial College London; 2016.
- [8] ATR. ATR 72-600 Performance Specification Sheet; [cited 10/06/2016]. Available from: http://www.ATR-aircraft.com/datas/download_center/27/15c2188_fiche_72_web_27.pdf.
- [9] Roskam J. Airplane Design - Part V: Component Weight Estimation. DARcorporation; 1985.
- [10] Torenbeek E. Synthesis of Subsonic Airplane Design. Springer; 1982.
- [11] Federal Aviation Administration. Aircraft Weight and Balance Control - FAA Advisory Circular 120-27E; 2005.
- [12] Roskam J. Airplane Design - Part VI: Preliminary Calculation of Aerodynamic, Thrust and Power Characteristics. DARcorporation; 1987.
- [13] Hoak DE, Finck RD, et al. USAF Stability and Control Data Compendium (DATCOM). Flight Control Division, Air Force Flight Dynamics Laboratory, WPAFB Ohio: United States Air Force; 1978. ADB072483.
- [14] Raymer DP. Aircraft Design: A Conceptual Approach. American Institute of Aeronautics and Astronautics; 1989.
- [15] Ng R. Hybrid Electric Regional Aircraft - Structural Design. Department of Aeronautics, Imperial College London; 2016.
- [16] Currey NS. Aircraft Landing Gear Design: Principles and Practices. American Institute of Aeronautics and Astronautics; 1988.
- [17] McCullagh J. Hybrid Electric Regional Aircraft - Structural Design. Department of Aeronautics, Imperial College London; 2016.
- [18] Babister AW. Aircraft Dynamic Stability and Response. Pergamon International; 1980.
- [19] Roskam J. Airplane Design - Part VII: Determination of Stability, Control and Performance Characteristics: FAR and Military Requirements. DARcorporation; 1986.
- [20] Malik F. Hybrid Electric Regional Aircraft - Propulsion and Systems. Department of Aeronautics, Imperial College London; 2016.
- [21] Pleho F. Hybrid Electric Regional Aircraft - Flight Testing. Department of Aeronautics, Imperial College London; 2016.
- [22] Tam N. Hybrid Electric Regional Aircraft - Flight Testing. Department of Aeronautics, Imperial College London; 2016.
- [23] Roskam J. Airplane Design - Part II: Preliminary Configuration Design and Integration of the Propulsion System. DARcorporation; 1985.
- [24] Roskam J. Airplane Design - Part III: Layout Design of Cockpit, Fuselage, Wing and Empennage: Cutaways and Inboard Profiles. DARcorporation; 1986.

A Phases of Flight Evaluated in the Constraint Diagram

The flight conditions evaluated in the constraint diagram are outlined in Table A.1 with their respective equations. FAR climb requirements can be found in FAR §25.113-25.121.

Table A.1: Flight Conditions Evaluated.

Takeoff	Runway Length = 1200m TOP = 95	$\frac{P}{W} _{TO} = \frac{V_{LOF}}{\eta} \frac{1}{C_{L,TO}TOP} \frac{W_{TO}}{S}$
Cruise	FL = 230; M = 0.55; n = 1 dh/dt = 0; dV/dt = 0;	$\frac{P}{W} _{TO} = \frac{V}{\eta_{prop}\alpha} \left(\frac{qC_{D0}}{W_{TO}/S} + \frac{1}{\pi Re} \frac{n^2\beta^2}{q} \frac{W_{TO}}{S} \right)$
30deg Turning Flight	FL = 230; M = 0.55; n = 1.15 dh/dt = 0; dV/dt = 0;	$\frac{P}{W} _{TO} = \frac{V}{\eta_{prop}\alpha} \left(\frac{qC_{D0}}{W_{TO}/S} + \frac{1}{\pi Re} \frac{n^2\beta^2}{q} \frac{W_{TO}}{S} \right)$
Service Ceiling	FL = 250 M = 0.5; n = 1 dh/dt = 100fpm	$\frac{P}{W} _{TO} = \frac{V}{\eta_{prop}\alpha} \left(\frac{qC_{D0}}{W_{TO}/S} + \frac{1}{\pi Re} \frac{n^2\beta^2}{q} \frac{W_{TO}}{S} \right) + \frac{\beta}{\alpha} \left(\frac{1}{V} \frac{dh}{dt} \right)$
Climb at BOC	FL = 0 V = 130 kts; n = 1 dh/dt = 1500fpm	$\frac{P}{W} _{TO} = \frac{V}{\eta_{prop}\alpha} \left(\frac{qC_{D0}}{W_{TO}/S} + \frac{1}{\pi Re} \frac{n^2\beta^2}{q} \frac{W_{TO}}{S} \right) + \frac{\beta}{\alpha} \left(\frac{1}{V} \frac{dh}{dt} \right)$

B Weight and Balance Detailed Systems Weights

A detailed break down of the weigh and balance of the individual components forming HERA.

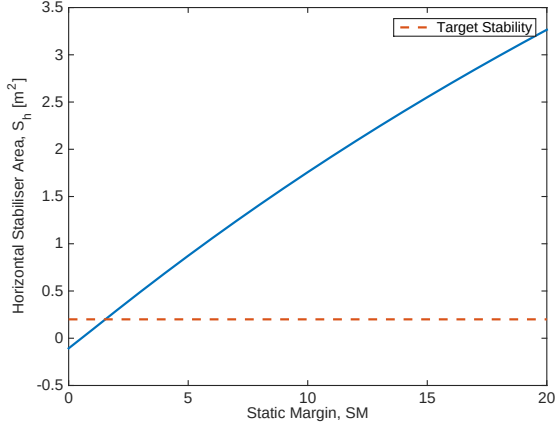
Table B.1: Class II Detailed Weight and Balance break down. X reference from the nose, Z reference from 1m below fuselage.

	W(kg)	X(m)	W*X (kg m)	Y (m)	W*Y (kg m)	Z (m)	W*Z (kg m)
FUSELAGE GROUP	3190	13.1	41797.7	0	0	2.3	7337
WING GROUP	1770	14.35	25399.5	0	0	3.5	6195
EMPENNAGE GROUP	300	28	8400	0	0	5.75	1725
ENGINE GROUP	1860.1	13.6	25297.9	0	0	3.5	6510.5
LANDING GEAR GROUP	1050	11.7	12321.8	0	0	1	1050
FIXED EQUIPMENT	3092.9	13.8	42763.1	0	0	2.6	8134.7
EMPTY WEIGHT	11263.0	13.8	155979.9	0	0	2.7	30952.2
TRAPPED FUEL/OIL	24	14.35	344.4	0	0	3.6	86.4
CREW	190.9	2.5	477.3	0	0	2.2	420
OPERATING EMPTY WEIGHT	11477.9	13.7	156801.6	0	0	2.7	31458.6
FUEL	2400	14.35	34440.0	0	0	3.6	8640
PASSENGERS	6708	12.2	81837.6	0	0	2.3	15428.4
BAGGAGE	829.1	20.7	17162.2	0	0	2.3	1865.5
MTOW	21415.0	13.6	290241.4	0	0	2.7	57392.5

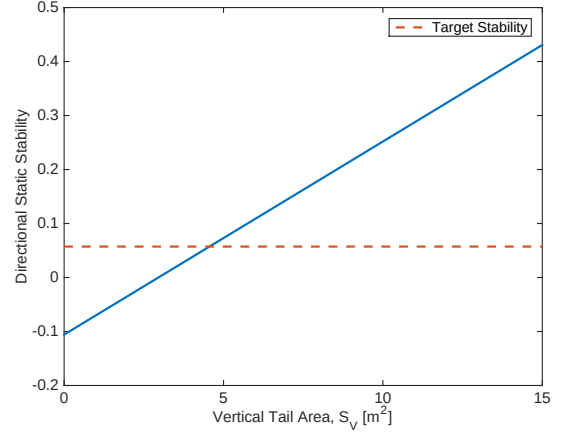
C Preliminary Tailplane Design

The preliminary tailplane sizing was conducted trying to obtain a value that minimally complied with stability requirements such that when performing the actual in depth analysis the results converged much more rapidly.

These methods are based on tail volume coefficients of similar past aircraft and tend to give a slightly smaller tail than that preferred by the HERA Flight Testing Division. Therefore, a preliminary longitudinal and directional static stability analysis was performed varying the tailplane area [23, 24]. These are shown on Figure C.1 for longitudinal and directional stability.



(a) Variation of preliminary Static Margin with Horizontal Stabiliser Area.



(b) Variation of preliminary directional stability with Vertical Tail Area.

Figure C.1: Preliminary Tailplane Sizing Method.

As seen, the resulting vertical and horizontal tailplane areas are small compared to the final value. However, they were satisfactory at providing an initial estimation point prior to the thorough stability analysis.